

ECS Project Step 3 – Create ECR Repository

To Begin with the Lab

Summary of the Lab

In this step, an **Amazon ECR** private repository was created for the PHP application image built on the Docker-enabled build machine. The repository was needed because **Amazon ECS** cannot build images itself and must pull pre-built images from a registry. During repository creation, the lab used the repository name `cloudfreeks-s3-php`, selected **Mutable** tag behavior for learning purposes, and kept the default **AES-256** encryption option. The repository was then created successfully and prepared for the image push in the next step.

Understanding Amazon ECR Repository and Tag Mutability

Amazon ECR is a private container registry used to store Docker images for services like **Amazon ECS**. It exists because ECS can run container images, but it does not build or store them, so images must first be pushed into a registry. Tag mutability controls whether the same image tag can be overwritten when a new version is pushed. For example, if `cloudfreeks-s3-php:latest` is pushed again after code changes, a mutable repository allows the newer image to replace the older one, while an immutable repository blocks that overwrite and helps prevent accidental deployment mistakes.

Example:

A developer builds version 1 of `cloudfreeks-s3-php` and pushes it with the tag `latest`. Later, after fixing the upload feature, version 2 is built and pushed with the same tag. In a mutable repository, the second push replaces the first image, but in an immutable repository, the old version stays protected.

Lab Steps

Step 1 – Open Amazon ECR

- Sign in to the **AWS Management Console**.
- Navigate to **Amazon ECR**.
- Click **Create repository**.

Containers

Amazon Elastic Container Registry

Share and deploy container software, publicly or privately

Amazon Elastic Container Registry (ECR) is a fully managed container registry that makes it easy to store, manage, share, and deploy your container images and artifacts anywhere.

How it works

Benefits

Create a repository

Create

Pricing (US)

You pay only for the amount of data you store in your public or private repositories and data transferred to the Internet.

[ECR pricing](#)

Enable private connection to ECR private repositories using VPC interface endpoints and reduce cost.

[Learn more](#)

[Deploy VPC interface endpoints](#)

Getting started

[What is Amazon ECR?](#)

https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1

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Step 2 – Create the Private Repository

- Select the private repository option.
- Enter the repository name: cloudfreeks-s3-php
- Keep the repository private because **Amazon ECR** supports private repositories for this project.
- Click **Create repository**.

Amazon ECR > Private registry > Repositories > Create private repository

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.

463646775279.dkr.ecr.us-east-1.amazonaws.com/cloudfreeks-s3-php

18 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./.

Image tag settings

Image tag mutability
Choose the tag mutability setting.

☒ **Mutable**
Image tags can be overwritten.

☐ **Immutable**
Image tags can't be overwritten.

Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.

[Add filter](#)

Filters must only contain letters, numbers, and special characters (-, _). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.

Encryption settings

Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

Warning: The encryption settings for a repository can't be changed once the repository is created.

CloudShell Feedback Console Mobile App

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Step 3 – Configure Tag Mutability

- Review the **Image tag mutability** setting.
- Understand that **Mutable** allows the same tag to be overwritten.
- Understand that **Immutable** prevents accidental tag replacement.

- Keep **Mutable** for this training project.

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
463646775279.dkr.ecr.us-east-1.amazonaws.com/cloudfreeks-s3-php
18 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./.

Image tag settings [info](#)

Image tag mutability

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 [Add filter](#)
Filters must only contain letters, numbers, and special characters (.,_-). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.

Encryption settings [info](#)

Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

Step 4 – Configure Encryption

- Review the encryption settings.
- Keep the default **AES-256 / ECR managed** encryption option.
- Note that **AWS KMS** can be used later for stronger compliance or custom key control.

Encryption settings [info](#)

Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

☒ **AES-256**
Amazon ECR managed Advanced Encryption Standard (AES) encryption

☐ **AWS KMS**
AWS Key Management Service (KMS)

Image scanning settings - deprecated

[Cancel](#) [Create](#)

Step 5 – Complete the Repository Creation

- Click **Create** to finish creating the repository.

Amazon ECR > Private registry > Repositories > Create private repository

Mutable
Image tags can be overwritten.

Immutable
Image tags can't be overwritten.

Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.
Add filter

Filters must only contain letters, numbers, and special characters (.,-). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.

Encryption settings Info

The encryption settings for a repository can't be changed once the repository is created.

Encryption configuration
Repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.

AES-256
Repositories use the industry standard Advanced Encryption Standard (AES) encryption.

AWS KMS
AWS Key Management Service (KMS)

Image scanning settings - deprecated

Cancel Create

- Review the generated repository URI.

Amazon ECR > Private registry > Repositories

Successfully created private repository, cloudfreeks-s3-php

Private repositories (10)

Search by repository substring

Repository name	URI	Created at	Tag immutability	Encryption type
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	July 08, 2026, 18:28:36 (UTC+05.5)	Mutable	AES-256
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	July 08, 2026, 19:10:09 (UTC+05.5)	Mutable	AES-256
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	January 01, 2026, 11:51:38 (UTC+05.5)	Mutable	AES-256
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	January 10, 2024, 22:05:12 (UTC+05.5)	Mutable	AES-256
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	May 30, 2026, 19:50:58 (UTC+05.5)	Mutable	AES-256
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	January 09, 2026, 18:37:09 (UTC+05.5)	Mutable	AES-256
cloudfreeks-s3-php	1.amazonaws.com/cloudfreeks-s3-php	June 02, 2026, 22:53:59 (UTC+05.5)	Mutable	AES-256

- Confirm that the repository is ready to receive the Docker image from the build machine.

Verification

- Verified that the **Amazon ECR** repository was created successfully.
- Verified that the repository name was set to **cloudfreeks-s3-php**.
- Verified that the repository was created as a private repository.
- Verified that **Mutable** tag behavior was selected.
- Verified that default **AES-256** encryption was used.
- Verified that the repository URI was generated and ready for the next image push.